

I MADE ANANDA RYAN VIRYAVAN

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Education

Bina Nusantara University, Undergraduate *Bachelor of Science in Computer Science* Sept 2023 - Dec 2027

Technical Skills

Languages: Python, Java, C++, SQL, JavaScript, TypeScript, Bash

Backend & APIs: Spring Boot, FastAPI, Node.js, REST APIs, JUnit, Cypress

AI/RAG & Data: LangChain, OpenAI APIs, ChromaDB, Vector Embeddings, Pandas, NumPy

Frontend: React, HTML/CSS, JavaScript, Responsive Web Design

Cloud & DevOps: AWS, Docker, Kubernetes, Helm, ArgoCD, Terraform, GitHub Actions, Jenkins

Monitoring & Infrastructure: Datadog, OPA/Rego, Linux, Git, CI/CD, Microservices

Experience

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Projects - View all of my projects on GitHub

waspadaGe.mpa: Real-time Global Disaster & Climate risk map | Next.js, Typescript, Tailwind CSS, Leaflet.js, USGS API, BMKG API

- Built a real-time seismic monitoring platform that fetches and visualizes live earthquake data from dual sources — **USGS (Global)** and **BMKG (Official)** — with interactive map rendering and magnitude-based filtering (All, Mag 4+, Mag 6+)..
- Engineered an advanced analysis suite including **Seismic Activity Heatmap**, **7-Day Time Travel Playback**, and a **Seismic Statistics Dashboard** to support in-depth disaster pattern analysis.
- Implemented multi-layer map visualization with toggleable overlays for **tectonic plates**, **active fault lines**, and **climate risk projections**, alongside depth-based and magnitude-based color modes.
- Designed a bilingual (ID/EN) minimalist UI with **Web Push Notifications** support and a live seismic event list, deployed on Vercel at v1.2.0.

GeoWatch Indonesia: Interactive Active Volcano & Geothermal Potential Map | Next.js, TypeScript, Tailwind CSS, Leaflet.js, PVMBG API

- Built an interactive geospatial platform that monitors and visualizes **active volcanoes across Indonesia**, displaying real-time alert status levels — Normal, Waspada, Siaga, and Awas — sourced from official PVMBG data.
- Engineered a dynamic map interface with clickable volcano markers that surface detailed **volcanic activity reports** per volcano, including current alert level and geothermal potential classification.
- Implemented a **live data refresh** mechanism and statistical summary dashboard showing the distribution of volcanoes across all four alert levels at a glance.
- Designed a bilingual-ready, minimalist UI focused on disaster awareness, deployed on Vercel as part of a broader Indonesia geospatial monitoring ecosystem alongside WaspadaGempa